

Multiple Choice (10 marks)

1. One limitation of food webs in representing an ecosystem is that they do not
 - (a) include producers in the ecosystem.
 - (b) include all species in the ecosystem.
 - (c) show consumers in the ecosystem.
 - (d) show predator-prey relationships in the ecosystem.
2. The correct order of a plant's stages of growth and development is
 - (a) young plant -> seed -> mature plant
 - (b) seed -> mature plant -> young plant
 - (c) seed -> young plant -> mature plant
 - (d) mature plant -> young plant -> seed
3. Which of the following is a similarity between x-ray waves and sound waves?
 - (a) Both transfer energy.
 - (b) Both require a vacuum.
 - (c) Both have the same speed.
 - (d) Both have the same frequency.
4. What are the products in the reaction shown below? $\text{HCl} + \text{NaOH} \rightarrow$
 - (a) $\text{Na} + \text{Cl}_2$
 - (b) $\text{NaCl} + \text{OH}$
 - (c) $\text{NaOH} + \text{Cl}$
 - (d) $\text{NaCl} + \text{H}_2\text{O}$
5. There is a limited supply of copper on Earth. How does this affect the way copper is used?
 - (a) Copper should be recycled.
 - (b) Copper costs less than other materials.
 - (c) Copper is in low demand.
 - (d) Copper does not make up commercial products.

True / False (10 marks)

Answer True or False

6. True or False: Sound travels faster in saltwater than in freshwater because saltwater absorbs heat.
7. True or False: Sonar equipment measures the time delay of reflected waves to determine the depth of the ocean.
8. True or False: Heat energy is transferred from a hot rock to cool water mainly through evaporation.
9. True or False: A plastic yogurt container is ideal for storing unknown liquids because it provides a barrier.
10. True or False: Clearing tropical rainforests leads to a decrease in biodiversity as many species lose their habitat.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of nuclear fusion in the Sun, focusing on the specific layer where it occurs and its significance to the Sun's energy production and overall structure.
12. Discuss the physical changes that occur during the digestion process. How do these changes impact the overall digestion and absorption of nutrients in the body?

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